Bubblesort

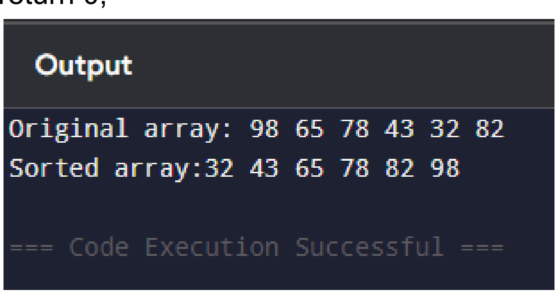
```
#include <stdio.h>
void bubblesort(int arr[], int n) {
    int temp;

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

int main() {
    int arr[] = {98, 65, 78, 43, 32, 82};
    int n = 6;
    int i;

    printf("Original array: ");
    for (i=0;i<6;i++)
    printf("%d ",arr[i]);


    bubblesort(arr, n);

    printf("\nSorted array:");
    for (i=0;i<6;i++)
    printf("%d ",arr[i]);


    return 0;
}
```

Output

```
Original array: 98 65 78 43 32 82
Sorted array:32 43 65 78 82 98


=== Code Execution Successful ===
```

Selection sort

```
#include<stdio.h>
void selectionsort(int arr[], int n) {
    int minindex, temp;

    for (int i = 0; i < n - 1; i++) {
        minindex = i;

        for (int j = i + 1; j < n; j++) {
            if (arr[j] < arr[minindex]) {
                minindex = j;
            }
        }

        temp = arr[i];
        arr[i] = arr[minindex];
        arr[minindex] = temp;
    }
}
int main() {
    int arr[] = {98, 65, 78, 43, 32, 82};
    int n = 6;
    int i;

    printf("Original array: ");{
    for (i=0;i<6;i++)
    printf("%d ",arr[i]);}


    selectionsort(arr, n);

    printf("\nSorted array:");{
    for (i=0;i<6;i++)
    printf("%d ",arr[i]);}


 return 0;
```




```
}
```

Insertion sort

```c
#include <stdio.h>

void insertionsort(int arr[], int n)
{
    int i, j, key;

    for (i = 1; i < n; i++)
    {
        key = arr[i];
        j = i - 1;

        while (j >= 0 && arr[j] > key)
        {
            arr[j + 1] = arr[j];
            j--;
        }

        arr[j + 1] = key;
    }
}

int main()
{
    int arr[] = {79, 90, 67, 52, 88};
    int n = 5;
    int i;

    printf("Original array: ");
    for (i = 0; i < n; i++)
        printf("%d ", arr[i]);

    insertionsort(arr, n);

    printf("\nSorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Output

```
Original array: 79 90 67 52 88
Sorted array: 52 67 79 88 90

=== Code Execution Successful ===
```